Ch 6.2

22:48 Mon, 7 Apr 84%

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The Fibonacci matrix is defined as:

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$$

Step 1: Eigenvalues of the Fibonacci Matrix

First, we need to find the eigenvalues of the matrix A. The characteristic equation is:

$$\det(A - \lambda I) = 0$$

where I is the identity matrix. The equation becomes:

$$\det\left(\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right) = 0$$

Simplifying this:

$$\det\left(\begin{pmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{pmatrix}\right) = 0$$

The determinant is:

$$(1-\lambda)(-\lambda) - 1 = 0$$

$$-\lambda + \lambda^2 - 1 = 0$$

$$\lambda^2 - \lambda - 1 = 0$$

Solving the quadratic equation for λ:

$$\lambda = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-1)}}{2(1)}$$

$$\lambda = \frac{1 \pm \sqrt{1+4}}{2}$$

$$\lambda = \frac{1 \pm \sqrt{5}}{2}$$

Thus, the eigenvalues are:

$$\lambda_1 = \frac{1+\sqrt{5}}{2}, \quad \lambda_2 = \frac{1-\sqrt{5}}{2}$$

Step 2: Diagonalization of t[illegible]onacci Matrix

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Step 2: Diagonalization of the Fibonacci Matrix

The diagonalization of the matrix A can be written as:

$$A = X\Lambda X^{-1}$$

where Λ is the diagonal matrix of eigenvalues:

$$\Lambda = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

The matrix X consists of the eigenvectors corresponding to λ_1 and λ_2.

The eigenvector for λ_1 satisfies:

$$A \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \lambda_1 \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

This gives the system of equations:

$$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \lambda_1 \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

Solving this system yields:

$$v_1 = \lambda_1 v_2$$

Thus, the eigenvector corresponding to λ_1 is:

$$\mathbf{v}_1 = \begin{pmatrix} \lambda_1 \\ 1 \end{pmatrix}$$

Similarly, for λ_2, the eigenvector is:

$$\mathbf{v}_2 = \begin{pmatrix} \lambda_2 \\ 1 \end{pmatrix}$$

Thus, the matrix X is:

$$X = \begin{pmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{pmatrix}$$

The inverse of X is:

$$X^{-1} = \frac{1}{\lambda_1 - \lambda_2} \begin{pmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{pmatrix}$$

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Similarly, for λ_2, the eigenvector is:

$$\mathbf{v}_2 = \begin{pmatrix} \lambda_2 \\ 1 \end{pmatrix}$$

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The inverse of X is:

$$X^{-1} = \frac{1}{\lambda_1 - \lambda_2} \begin{pmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{pmatrix}$$

Step 3: Calculating $X\Lambda^k X^{-1}$

Now we can calculate $X\Lambda^k X^{-1}$ for any k. The matrix Λ^k is:

$$\Lambda^k = \begin{pmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{pmatrix}$$

Multiplying these matrices:

$$X\Lambda^k X^{-1} = \begin{pmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{pmatrix} \frac{1}{\lambda_1 - \lambda_2} \begin{pmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{pmatrix}$$

Step 4: Second Component and Fibonacci Number

The second component of the result is the Fibonacci number F_k, given by:

$$F_k = \frac{\lambda_1^k - \lambda_2^k}{\lambda_1 - \lambda_2}$$

This is the closed-form expression for the k-th Fibonacci number, also known as Binet's formula.

Thus, F_k is calculated by:

$$F_k = \frac{\lambda_1^k - \lambda_2^k}{\lambda_1 - \lambda_2}$$

where $\lambda_1 = \frac{1+\sqrt{5}}{2}$ and $\lambda_2 = \frac{1-\sqrt{5}}{2}$.

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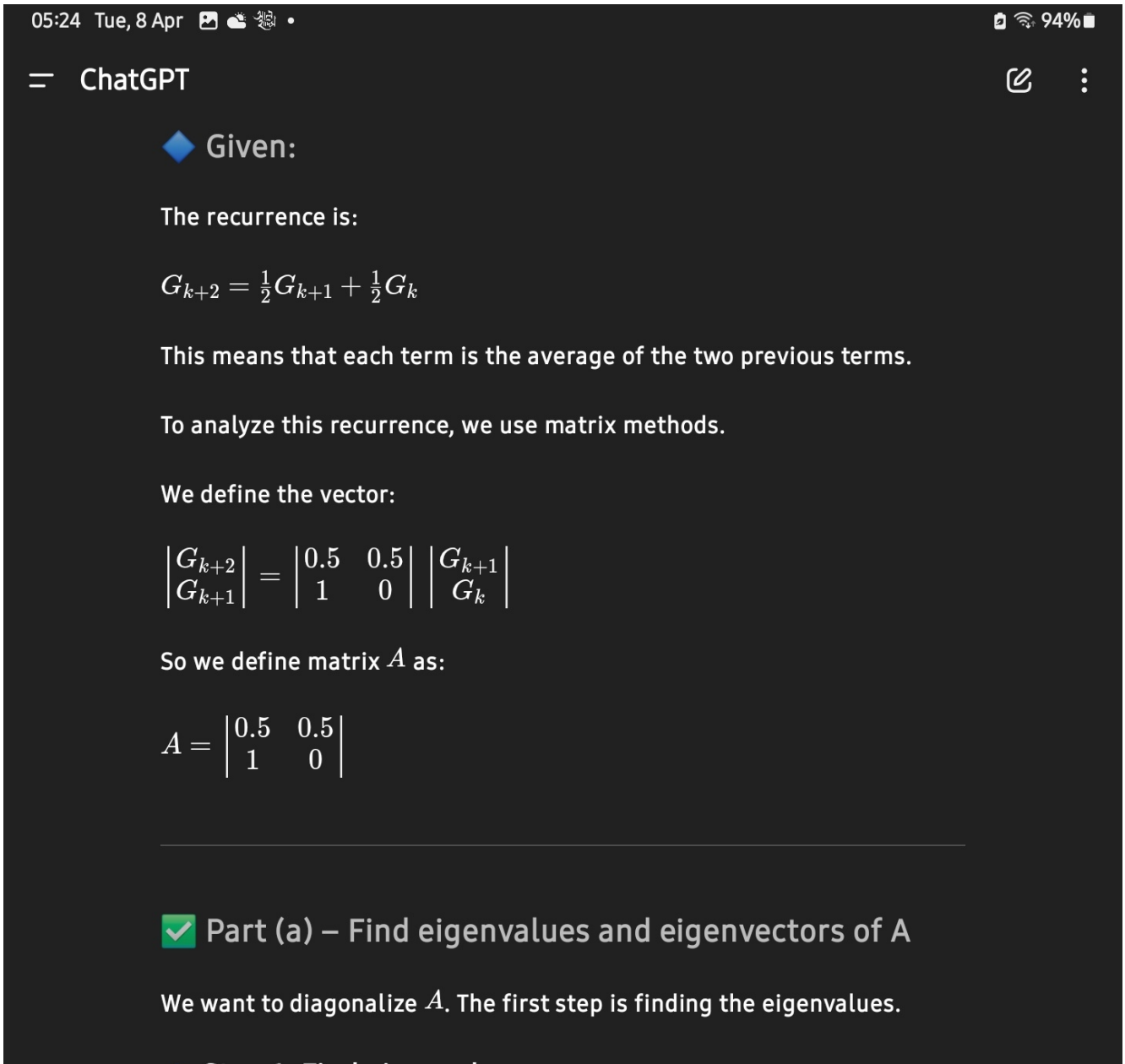
The matrix A is derived from the recurrence relation $G_{k+2} = \frac{1}{2}(G_{k+1} + G_k)$. To express this as a matrix equation:

$$\begin{bmatrix} G_{k+2} \\ G_{k+1} \end{bmatrix} = A \begin{bmatrix} G_{k+1} \\ G_k \end{bmatrix}$$

1. First row of A:
   $G_{k+2} = \frac{1}{2} G_{k+1} + \frac{1}{2} G_k$, so the coefficients are $[0.5, 0.5]$.
2. Second row of A:
   $G_{k+1} = 1 \cdot G_{k+1} + 0 \cdot G_k$, so the coefficients are $[1, 0]$.

Thus, the correct matrix is:

$$A = \begin{bmatrix} 0.5 & 0.5 \\ 1 & 0 \end{bmatrix}$$



🔷 Step 1: Find eigenvalues

Eigenvalues are the roots of the characteristic polynomial:

$$\det(A - \lambda I) = 0$$

$$\det\left(\begin{vmatrix} 0.5 - \lambda & 0.5 \\ 1 & -\lambda \end{vmatrix}\right) = (0.5 - \lambda)(-\lambda) - (0.5)(1) = 0$$

Now simplify:

$$-0.5\lambda + \lambda^2 - 0.5 = 0 \Rightarrow \lambda^2 - 0.5\lambda - 0.5 = 0$$

Solve using quadratic formula:

$$\lambda = \frac{0.5 \pm \sqrt{(0.5)^2 + 4(0.5)}}{2} = \frac{0.5 \pm \sqrt{2.25}}{2} = \frac{0.5 \pm 1.5}{2}$$

So, eigenvalues:

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05:25 Tue, 8 Apr • 94%
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So, eigenvalues:

- $\lambda_1 = \frac{0.5+1.5}{2} = 1$
- $\lambda_2 = \frac{0.5-1.5}{2} = -0.5$

🔷 Step 2: Find eigenvectors

For $\lambda_1 = 1$:

We solve:

$$(A - I)x = 0 \Rightarrow \left(\begin{vmatrix} 0.5 & 0.5 \\ 1 & 0 \end{vmatrix} - \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}\right) = \begin{vmatrix} -0.5 & 0.5 \\ 1 & -1 \end{vmatrix}$$

Solve:

$$\begin{vmatrix} -0.5 & 0.5 \\ 1 & -1 \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \end{vmatrix} = 0 \Rightarrow x_1 = x_2$$

So an eigenvector is:

$$x_1 = \begin{vmatrix} 1 \\ 1 \end{vmatrix}$$

For $\lambda_2 = -0.5$:

Now:

$$(A + 0.5I)x = 0 \Rightarrow \begin{vmatrix} 1 & 0.5 \\ 1 & 0.5 \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \end{vmatrix} = 0 \Rightarrow x_1 = -0.5x_2$$

So an eigenvector is:

$$x_2 = \begin{vmatrix} 1 \\ -2 \end{vmatrix}$$

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✅ Part (b) – Limit of $A^n = X\Lambda^n X^{-1}$ as $n \to \infty$

Now that we have:

- Eigenvalues: $\lambda_1 = 1$, $\lambda_2 = -0.5$
- Eigenvectors:

$$X = \begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} \quad \text{and} \quad \Lambda = \begin{vmatrix} 1 & 0 \\ 0 & -0.5 \end{vmatrix}$$

We write:

$$A^n = X\Lambda^n X^{-1}$$

🔷 What happens as $n \to \infty$?

$$\Lambda^n = \begin{vmatrix} 1^n & 0 \\ 0 & (-0.5)^n \end{vmatrix} \to \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} \text{ because } (-0.5)^n \to 0$$

Then:

$$A^\infty = X \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} X^{-1}$$

Compute this product. The final result (from the book) is:

$$A^{\infty} = \begin{vmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{vmatrix}$$

This is a projection matrix — it projects any vector onto the direction of the dominant eigenvector.

✅ Part (c) – Show that the sequence approaches $\frac{2}{3}$

We are given:

- $G_0 = 0$
- $G_1 = 1$

Let's write the state vector:

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- $G_0 = 0$
- $G_1 = 1$

Let's write the state vector:

$$\begin{vmatrix} G_1 \\ G_0 \end{vmatrix} = \begin{vmatrix} 1 \\ 0 \end{vmatrix}$$

Now apply A^n to this vector:

$$\begin{vmatrix} G_{n+1} \\ G_n \end{vmatrix} = A^n \begin{vmatrix} 1 \\ 0 \end{vmatrix}$$

As $n \to \infty$, this becomes:

$$A^{\infty} \begin{vmatrix} 1 \\ 0 \end{vmatrix} = \begin{vmatrix} \frac{2}{3} \\ \frac{2}{3} \end{vmatrix}$$

So both G_n and $G_{n+1} \to \frac{2}{3}$

Final Conclusion

The Gibonacci sequence defined by:

$$G_{k+2} = \frac{1}{2}G_{k+1} + \frac{1}{2}G_k, \quad G_0 = 0, G_1 = 1$$

Converges to $\frac{2}{3}$ as $k \to \infty$.
This is shown using:

- Eigenvalues/eigenvectors of the transition matrix
- Diagonalization of A
- Matrix limits

Would you like me to animate this concept visually or give a code simulation to see it in action?

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10

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Step 3: Conclusion

From the sequence above, we see that every third Fibonacci number (i.e.,) is congruent to 0 modulo 2, which means every third Fibonacci number is even.

Therefore, we have proved that every third Fibonacci number is even.

10 The rule Fk+2 = Fk+1 + Fk produces the pattern: even, odd, odd, even, odd, odd, .

Yes, the rule $F_{k+2} = F_{k+1} + F_k$ generates the Fibonacci sequence, and when we look at the sequence modulo 2 (even or odd), we observe the following pattern:

Starting from $F_0 = 0$ and $F_1 = 1$, we compute the Fibonacci sequence

modulo 2:

$$
\begin{aligned}
&F_0 = 0 \quad (\text{even}) \\
&F_1 = 1 \quad (\text{odd}) \\
&F_2 = F_1 + F_0 = 1 + 0 = 1 \quad (\text{odd}) \\
&F_3 = F_2 + F_1 = 1 + 1 = 2 \equiv 0 \pmod 2 \quad (\text{even}) \\
&F_4 = F_3 + F_2 = 0 + 1 = 1 \quad (\text{odd}) \\
&F_5 = F_4 + F_3 = 1 + 0 = 1 \quad (\text{odd}) \\
&F_6 = F_5 + F_4 = 1 + 1 = 2 \equiv 0 \pmod 2 \quad (\text{even}) \\
&F_7 = F_6 + F_5 = 0 + 1 = 1 \quad (\text{odd}) \\
&F_8 = F_7 + F_6 = 1 + 0 = 1 \quad (\text{odd}) \\
&F_9 = F_8 + F_7 = 1 + 1 = 2 \equiv 0 \pmod 2 \quad (\text{even})
\end{aligned}
$$

Pattern Observed

The Fibonacci sequence modulo 2 follows a repeating cycle of:

$$\text{even, odd, odd, even, odd, odd, even, odd, odd, even, ...}$$

Conclusion

The rule $F_{k+2} = F_{k+1} + F_k$ produces the pattern: even, odd, odd, even, odd, odd, ..., and we see that every third Fibonacci number is even.

11 True or false: If the eigenvalues of A are 2, 2, 5 then the matrix is certainly

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11

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11. Eigenvalues of matrix A are 2, 2, and 5. Is the matrix:

(a) Invertible?

A matrix is invertible if none of its eigenvalues are zero. Given that the eigenvalues are 2, 2, and 5, none of them are zero. Therefore, the matrix is invertible.

Answer: True.

(b) Diagonalizable?

(b) Diagonalizable?

For a matrix to be diagonalizable, it must have enough linearly independent eigenvectors to form a basis for the vector space.

- The matrix has eigenvalues 2 (which is repeated) and 5.
- For the matrix to be diagonalizable, there must be 3 linearly independent eigenvectors corresponding to these eigenvalues.

If the algebraic multiplicity of the eigenvalue 2 is 2 (i.e., it repeats twice), we need 2 linearly independent eigenvectors associated with the eigenvalue 2. If the matrix provides 2 independent eigenvectors for eigenvalue 2, and 1 for eigenvalue 5, the matrix is diagonalizable.

Since there is no additional information about the eigenvectors of A, we assume that the matrix has enough independent eigenvectors (i.e., it is diagonalizable).

~~Answer: True.~~ False (not certain)

(c) Not diagonalizable?

For the matrix to be not diagonalizable, there must not be enough linearly independent eigenvectors. As mentioned, if the matrix has 2 independent eigenvectors for the eigenvalue 2 and 1 for the eigenvalue 5, it is diagonalizable. Therefore, it is not guaranteed to be non-diagonalizable.

Answer: False.

12. If the only eigenvectors of A are multiples of $(1, 4)$, then A has:

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12

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For the matrix to be not diagonalizable, there must not be enough linearly independent eigenvectors. As mentioned, if the matrix has 2 independent eigenvectors for the eigenvalue 2 and 1 for the eigenvalue 5, it is diagonalizable. Therefore, it is not guaranteed to be non-diagonalizable.

Answer: False.

12. If the only eigenvectors of A are multiples of $(1, 4)$, then A has:

(a) No inverse?

If the only eigenvectors of A are multiples of $(1, 4)$, this suggests that A has a single eigenvalue with algebraic multiplicity greater than 1 (since there is only one linearly independent eigenvector). If a matrix has fewer linearly independent eigenvectors than its size, it is not diagonalizable. However, the matrix might still have an inverse if its eigenvalue is non-zero.

Without knowing the exact eigenvalue, we cannot conclude that A has no inverse just from the eigenvectors. For A to not be invertible, its eigenvalue must be zero.

Answer: False.

(b) A repeated eigenvalue?

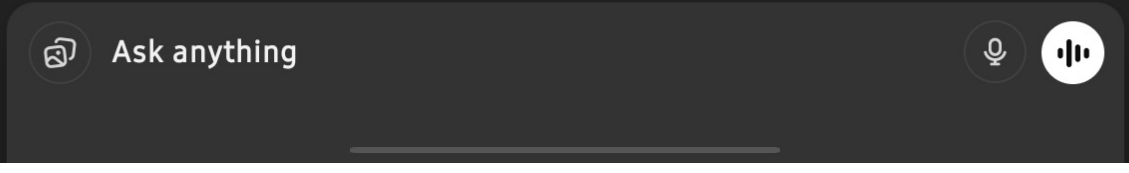
Since the only eigenvectors of A are multiples of $(1, 4)$, this implies that A has a repeated eigenvalue (the eigenvalue corresponding to the eigenvector $(1, 4)$ is likely repeated).

Answer: True.

(c) No diagonalization?

Since A has only one linearly independent eigenvector corresponding to a repeated eigenvalue, it does not have enough linearly independent eigenvectors to form a basis for diagonalization. Therefore, the matrix cannot be diagonalized.

Answer: True.



(a) True (All eigenvalues are non-zero $\Rightarrow$ invertible).
(b) False (Repeated eigenvalues do not guarantee diagonalizability).
(c) False (Not necessarily non-diagonalizable).

Question 12:

(a) False (No information about eigenvalues being zero).

(b) True (Repeated eigenvalue with geometric multiplicity 1).

(c) True (Insufficient eigenvectors for diagonalization).

(13) $A = \begin{bmatrix} 8 & 3 \\ -3 & 2 \end{bmatrix}$ $A = \begin{bmatrix} 9 & 4 \\ -4 & 1 \end{bmatrix}$ $A - 5I = \begin{bmatrix} 4 & 4 \\ -4 & -4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

$\det = 25$

$\lambda = 5$ repeated

$x_1 + x_2 = 0$

$\therefore \hat{x} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$A = \begin{bmatrix} 10 & 5 \\ -5 & 0 \end{bmatrix}$ only eigvectors $x = (c, -c)$

(14) $A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ any of these changed makes it diagonalizable.

$0.6 \times 0.1 = 0.06 - 0.36$

(15) $\lambda^2 - .7\lambda - 0.3$

$A_1 \begin{bmatrix} .6 & .3 \\ .4 & .1 \end{bmatrix}$ $\lambda_1 = 1, \lambda_2 = -0.3$

so $\lambda_{max} = 1$

no all $\lambda > 1$

So $A^K \not\to 0$

$A^K \to A^\infty$

$A_2 = \begin{bmatrix} .6 & .9 \\ .1 & .6 \end{bmatrix}$

$\det(A_2) = \lambda^2 - 1.2\lambda + 0.27$

$\lambda_1 = 0.9, \quad \lambda_2 = 0.3$

$0.6 \pm 0.3 \rightarrow 0.9, \ 0.3$ (Same)

$\lambda < 1$

$A_2^k \rightarrow 0 \quad \text{as } k \rightarrow \infty$

(16) $\Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 0.3 \end{bmatrix} \quad \Lambda^\infty = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

$A^\infty = X \Lambda^\infty X^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$ steady

06:48 Tue, 8 Apr • 91%

Matrix Derivation for Recurrence Relation Analysis

Inverse of X is 1/2 [1 1; 1 -1]. So when they compute the product, they get [1 0; 1 0] multiplied by 1/2 [1 1; 1 -1], which gives 1/2 [1 1; 1 1]. This matrix averages the two components of any vector, effectively projecting it onto the line spanned by [1, 1]. Since this projection doesn't change with further applications of , it's a steady state.

Wait, but if you apply again, it should give the same result. Let me verify by squaring . If , then . So yes, applying it multiple times doesn't change it—it's idempotent, which is a property of projection matrices. Hence, once you reach , further applications don't alter the result, hence "steady".

Therefore, the term "steady-state" comes from the fact that the system no longer changes once it reaches this state, stabilized by the dominant eigenvalue's eigenvector.

The matrix A_∞ is a steady-state matrix because it represents a projection onto the dominant eigenvector associated with the eigenvalue 1. Here's why:

1. Eigenvalue Behavior:
   - The eigenvalue 1 remains unchanged with each power A_1^k, while the eigenvalue 0.2 vanishes (\(0.2^k \to 0 \).
   - This means only the eigenvector corresponding to $\lambda = 1$ (i.e., [1, 1]^T\)) persists as \(k \to \infty .
2. Projection Property:
   - A_∞ projects any vector onto the subspace spanned by $[1, 1]^T$. For example:

$$A_\infty \begin{bmatrix} a \\ b \end{bmatrix} = \frac{1}{2} \begin{bmatrix} a+b \\ a+b \end{bmatrix},$$

     which averages the components and aligns the result with $[1, 1]^T$.
   - Further applications of A_∞ leave the result unchanged (\(A_\infty^2 = A_\infty \), confirming its steady nature.
3. Idempotence:
   - A_∞ is idempotent: $A_\infty^k = A_\infty$ for all $k \geq 1$. This means the system stabilizes and no longer evolves after reaching A_∞.

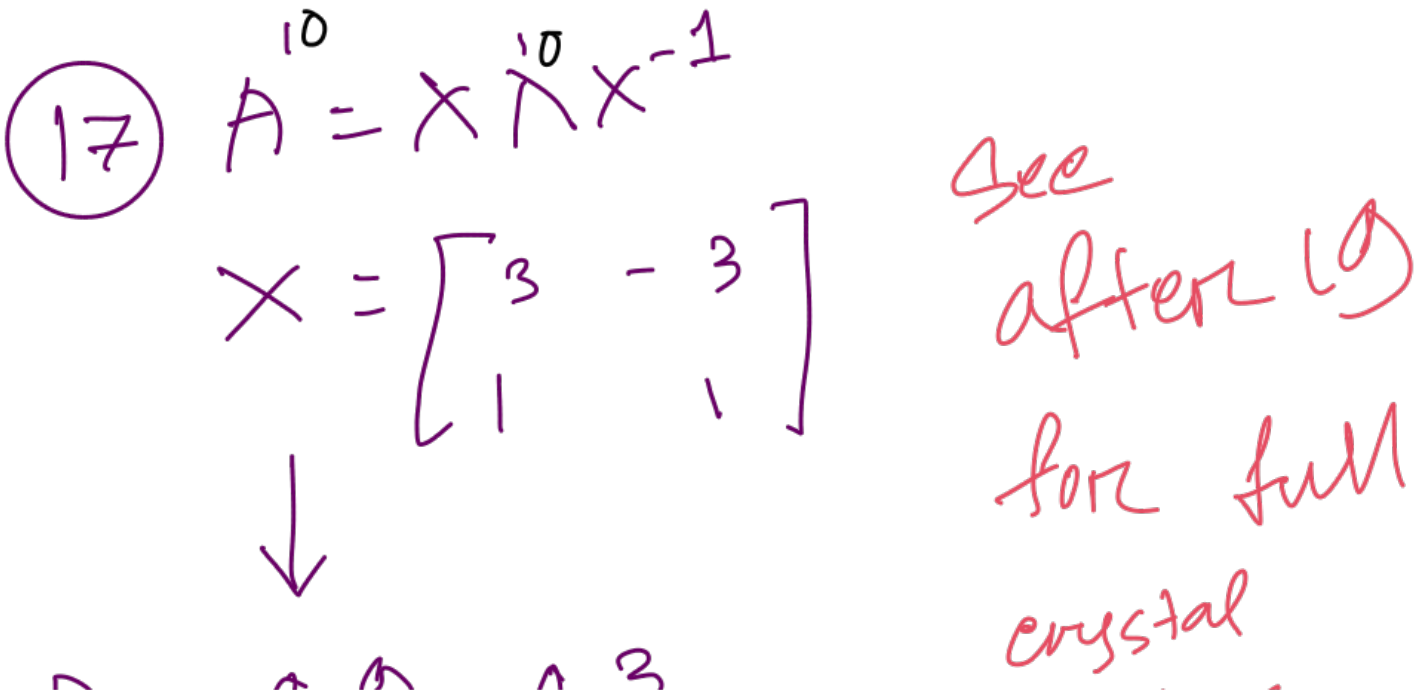
Thus, A_∞ is "steady" because it captures the long-term behavior of the system, filtering out transient components and retaining only the dominant eigenvector's contribution.

It projects any vector onto the dominant eigenvector, which remains unchanged under fu

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$\lambda = 0.3, 0.9$ clear.

$$A = \begin{bmatrix} .6 & .9 \\ .1 & -.6 \end{bmatrix}$$

$\lambda = 0.9$

$$\begin{bmatrix} -0.3 & 0.9 \\ 0.1 & -0.3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$0.3x_1 = 0.9x_2$$

$$x_1 = 3x_2$$

$$x_1 + 3x_2 = 0$$

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$x_1 = 3x_2$$

$\lambda = 0.3$

$$\begin{bmatrix} 0.3 & 0.9 \\ .1 & 0.3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = 0$$

$$0.3y = -0.9y_2$$

$$y_1 = -3y_1 \qquad \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

$$X = \begin{bmatrix} 3 & 3 \\ 1 & -1 \end{bmatrix}$$

$$X^{-1} = \frac{1}{-6}\begin{bmatrix} -1 & -3 \\ -1 & 3 \end{bmatrix}$$

$$A = \begin{bmatrix} 0.7 & 0 \\ 0 & 0.3 \end{bmatrix}^{10}$$

$$= \begin{bmatrix} 0.35 & 0 \\ 0 & 5.9\times10^{-6} \end{bmatrix}$$

(18)

$$\begin{bmatrix} 1 & 0 \\ 0 & 3^k \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 \\ \end{bmatrix}$$

$$\begin{bmatrix} -3^k & 3^k \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -3^k & 3^k \end{bmatrix}$$

$$\begin{bmatrix} 1+3^k & 1-3^k \\ 1-3^k & 1+3^k \end{bmatrix}$$ ✓

(19) $B = \begin{bmatrix} 5 & 1 \\ 0 & 4 \end{bmatrix}$

$\lambda = 5, 4$

$\lambda = 5,$

$$\begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \therefore x_2 = 0$$

$$V_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$\lambda = 4$

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 \end{bmatrix} \begin{pmatrix} x_1 \end{pmatrix}$$

$$\cdot v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \qquad X^{-1} = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$$

$$K = \begin{bmatrix} 5 & 0 \\ 0 & 4 \end{bmatrix}^k$$

$$= \begin{bmatrix} 5^k & 0 \\ 0 & 4^k \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 5^k & 5^k \\ 0 & -4^k \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 5^k & 5^k \\ 0 & -4^k \end{bmatrix}$$

$$= \begin{bmatrix} 5^k & 5^k - 4^k \\ 0 & 4^k \end{bmatrix}$$

(20) $A = X \Lambda X^{-1}$

$$\det(A) = \det(X)\det(\Lambda)\det(X^{-1})$$

$$= \det(\Lambda)$$

$$= \lambda_1 \lambda_2 \lambda_3 \cdots \lambda_n$$

(Only when A is diagonizable)

(21)

1. **Trace is Invariant under Cyclic Permutation**

The trace of the product of two matrices X and Y is invariant under cyclic permutation. That is, `trace(XY) = trace(YX)`.

Let:

X = [a b]
    [c d]

Y = [q r]
    [s t]

Now compute:

- **XY**:
  XY = [aq + bs ar + bt]
       [cq + ds cr + dt]
  trace(XY) = aq + bs + cr + dt

- **YX**:
  YX = [qa + rc qb + rd]
       [sa + tc sb + td]
  trace(YX) = qa + rc + sb + td

Clearly, `trace(XY) = trace(YX)`.

This works because the trace operation is **cyclic**, meaning `trace(AB) = trace(BA)`.

---

2. **Diagonalizable Matrices: Trace is Sum of

Eigenvalues**

If $A = X \Lambda X^{-1}$ is diagonalizable (where Λ is a diagonal matrix of eigenvalues λ_i), then:

- $\text{trace}(A) = \text{trace}(X \Lambda X^{-1})$

By the cyclic property of trace:

- $\text{trace}(X \Lambda X^{-1}) = \text{trace}(\Lambda X^{-1} X) = \text{trace}(\Lambda)$

Since Λ is diagonal, its trace is simply the sum of its diagonal elements (the eigenvalues):

- $\text{trace}(A) = \text{sum}(\lambda_i)$

Thus, the trace of A is the sum of its eigenvalues.

---

3. **Why AB - BA = I is Impossible**

Let's assume $AB - BA = I$. Taking the trace of both sides:

- $\text{trace}(AB - BA) = \text{trace}(AB) - \text{trace}(BA) = 0$ (since $\text{trace}(AB) = \text{trace}(BA)$)

But the trace of the identity matrix I is n, where I is an $n \times n$ matrix:

- $\text{trace}(I) = n$

This gives us:

- `0 ≠ n`

This is a contradiction, so `AB - BA = I` is impossible.

Summary

Statement	Key Reason
`trace(XY) = trace(YX)`	Cyclic property of trace
`trace(A) = sum(λᵢ)`	Diagonalizable matrices' trace is sum of eigenvalues
`AB - BA = I` is impossible	Trace of `AB - BA` is 0, while trace of `I` is non-zero

(22) $A = X\Lambda X^{-1}$, $B = \begin{bmatrix} A & 0 \\ 0 & 2A \end{bmatrix}$

$= \begin{bmatrix} X & 0 \\ 0 & X \end{bmatrix} \begin{bmatrix} \Lambda & 0 \\ 0 & 2\Lambda \end{bmatrix} \begin{bmatrix} X^{-1} & 0 \\ 0 & X^{-1} \end{bmatrix}$

So $\lambda_B = \lambda_1 \dots \lambda_n$ from A.

additional, $2\lambda_1 \dots 2\lambda_n$ from 2A

23

07:41 Tue, 8 Apr • 87%

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We are considering all 4×4 matrices A such that:

$$A = X\Lambda X^{-1}$$

for some diagonal matrix Λ, and fixed matrix X.

So, all matrices diagonalized by the same X.

We want to show:

1. This set forms a subspace.
2. Determine this subspace when $X = I$.
3. Find its dimension.

✅ Step 1: Show It's a Subspace

Let's define the set:

$$\mathcal{S} = \{A \in \mathbb{R}^{4\times 4} \mid A = X\Lambda X^{-1}, \text{ for some diagonal } \Lambda\}$$

Let's show that $\mathcal{S}$ is a subspace of $\mathbb{R}^{4\times 4}$:

🔸 Closure under addition

Let $A_1 = X\Lambda_1 X^{-1}$ and $A_2 = X\Lambda_2 X^{-1}$. Then:

$$A_1 + A_2 = X\Lambda_1 X^{-1} + X\Lambda_2 X^{-1} = X(\Lambda_1 + \Lambda_2)X^{-1}$$

Since $\Lambda_1 + \Lambda_2$ is diagonal, $A_1 + A_2 \in \mathcal{S}$

🔸 Closure under scalar multiplication

Let $c \in \mathbb{R}$, then:

$$cA_1 = c(X\Lambda_1 X^{-1}) = X(c\Lambda_1)X^{-1}$$

And $c\Lambda_1$ is diagonal, so $cA_1 \in \mathcal{S}$

✅ So $\mathcal{S}$ is a subspace of $\mathbb{R}^{4\times 4}$

Ask anything

07:41 Tue, 8 Apr
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✅ Step 2: When $X = I$

If $X = I$, then:

$$A = X\Lambda X^{-1} = I\Lambda I = \Lambda$$

So A must be a diagonal matrix.

Thus, in this case:

$$\boxed{\mathcal{S} = \text{the space of all diagonal } 4 \times 4 \text{ matrices}}$$

✅ Step 3: What is the Dimension?

A 4×4 diagonal matrix has non-zero entries only on the diagonal, and there are 4 of them.

So:

$$\boxed{\text{Dimension} = 4}$$

Because the basis is:

$$\left\{ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \right\}$$

🎯 Summary:

- The set of matrices diagonalized by the same X forms a subspace of $\mathbb{R}^{4\times4}$
- When $X = I$, that subspace is the space of diagonal matrices
- Dimension of that subspace is

Ask anything

(24) $A^2 = A$

$A(Ax) = Ax$ indempotent Mat.

If A has columns x1, . . . , xn then column by column, A2 = A means every Axi = xi.All vectors in the column space (combinations of those columns xi) are eigenvectorswith λ = 1. Always the nullspace has λ = 0 (A might have dependent columns,so there could be less than n eigenvectors with λ = 1). Dimensions of those spacesC(A) and N(A) add to n by the Fundamental Theorem, so A is diagonalizable(n independent eigenvectors altogether).

$A\vec{v} = \lambda\vec{v}$

$A^2\vec{v} = \lambda^2\vec{v}$ Also $A^2\vec{v} = \lambda\vec{v}$

$\therefore \lambda^2\vec{v} = \lambda\vec{v}$

$\therefore (\lambda^2 - \lambda)\vec{v} = 0$

$\therefore \lambda(\lambda - 1) = 0$ $\lambda = 0, 1$

↓ Null ↓ col

$R^n = col(A) + N(A)$

always diagonizable for indempotent.

(25) Enough subspace dimension vs enough eigvec

N(A) + C(A) overlap

C(A) might not have n indepe eigvector.

for details check self ch

Soln 6.2 #(25) note

(26) $\lambda = 9, -1$

for $\sqrt{B} = X\sqrt{\Lambda}X^{-1}$

so $\sqrt{9} + \sqrt{-1}$ trace

imaginary

08:28 Tue, 8 Apr • 84%

Matrix Factorization into XΛX^{-1} Explained

Explanation and Analysis

The key distinction lies in the structure and eigenvalues of the matrices:

1. Matrix $-I = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$:

- Eigenvalues: Both are -1 (repeated).
- Real Square Root Exists: ✅ Yes.
  - The matrix $R = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ (a 90° rotation matrix) satisfies $R^2 = -I$.
  - Despite having negative eigenvalues, $-I$ is a scalar multiple of the identity matrix. This symmetry allows a non-diagonal real matrix (like \(R \) to act as a square root.
  - Geometrically, R represents a rotation, and squaring it produces a 180° rotation (\(-I \)).

2. Matrix $B = \begin{bmatrix} 4 & 5 \\ 5 & 4 \end{bmatrix}$:

- Eigenvalues: 9 (positive) and -1 (negative).
- Real Square Root Exists: ❌ No.

ANIMA

- A real square root would require eigenvalues ± 3 (for \sqrt{9} \)) and \(\pm i (for \(\sqrt{-1} \).
- The eigenvalue -1 forces imaginary components in the square root matrix.
- Unlike $-I$, B is not a scalar multiple of the identity matrix and lacks symmetry to "hide" the imaginary terms.

Key Takeaways:

- Negative eigenvalues can sometimes have real square roots if the matrix structure compensates (e.g., \(-I \).
- Mixed eigenvalues (positive and negative) require complex entries in the square root, making a real solution impossible.

Matrix	Eigenvalues	Real Square Root?	Reason
$-I$	$-1, -1$	✅Yes	Symmetric scalar matrix allows non-diagonal real roots.
B	$9, -1$	❌No	Negative eigenvalue forces imaginary entries.

Final Answer:

- $-I$ has a real square root because its structure (scalar identity matrix) compensates for negative eigenvalues.
- B lacks a real square root due to mixed eigenvalues (9 and \(-1\), requiring complex entries.

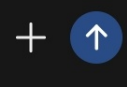
$$\boxed{\begin{array}{l}\text{Matrix } -I \text{ has a real square root (e.g., } R=\begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix}).\\ \text{Matrix } B \text{ does not have a real square root.}\end{array}}$$



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(28) $A = X\Lambda_1 X^{-1}$, $B = X\Lambda_2 X^{-1}$

$$AB = X\Lambda_1 X^{-1} X \Lambda_2 X^{-1}$$

$$= X \Lambda_1 \Lambda_2 X^{-1}$$

$$= X \Lambda_2 \Lambda_1 X^{-1}$$

$$= X \Lambda_2 X^{-1} X \Lambda_1 X^{-1} = BA$$

(29) $\begin{bmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{bmatrix}$ $= \det(A - \lambda I) = 0$

$$(1-\lambda)(-\lambda) - 1$$

$$= \lambda^2 - \lambda - 1 = 0$$

So $A^2 - A - I = 0$

(30) check self ch notes 6.2#30

(31) $A = \begin{bmatrix} 0 & \\ 0 & 2 \end{bmatrix}$ $= B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$AB = \begin{pmatrix} a & b \\ 2c & 2d \end{pmatrix}$$

$$BA = \begin{pmatrix} a & 2b \\ c & 2d \end{pmatrix}$$

$\therefore b = 2b \quad \therefore b = 0$

$2c = c \quad \therefore c = 0$

$$\therefore B = \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix}$$

$$AB - BA = \begin{bmatrix} 0 & -b \\ c & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

09:08 Tue, 8 Apr • 81%

Matrix Factorization into XΛX^{-1} Explained

Step 1: Derive Equations from $AB = BA$

Given:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

compute $AB - BA$:

$$AB - BA = \begin{pmatrix} 0 & -b \\ c & 0 \end{pmatrix}.$$

Setting $AB - BA = 0$ gives:

1. $0 = 0$ (from entry (1,1)),
2. $-b = 0$,
3. $c = 0$,
4. $0 = 0$ (from entry (2,2)).

Step 2: Write the System as $M\mathbf{x} = 0$

Let $\mathbf{x} = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$. The equations translate to:

$$\begin{cases} 0a - 1b + 0c + 0d = 0 & (\text{from } -b = 0), \\ 0a + 0b + 1c + 0d = 0 & (\text{from } c = 0). \end{cases}$$

The trivial equations $0 = 0$ contribute no constraints. The coefficient matrix M is:

is:

$$M = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Step 3: Compute the Rank

The rank of M is the number of linearly independent rows. Here:

- Non-zero rows: First and second rows.
- Zero rows: Third and fourth rows (no contribution).

Thus, the rank of the coefficient matrix is 2.

Conclusion

- B must be diagonal: $B = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}$.
- The rank of the coefficient matrix is $\boxed{2}$.

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09:14 Tue, 8 Apr • 81%

Matrix Derivation for Recurrence Relation Analysis

> e1 = [1, 0]^T and e2 = [0, 1]^T, respectively.
>
> Now, matrix B is a 2x2 matrix that commutes with A, meaning AB = BA. From the previous problem, we determined that B must also be diagonal, so B has the form [[a, 0], [0, d]]. Diagonal matrices have their eigenvalues on the diagonal, so the eigenvalues of B are a and d. The eigenvectors for B would also be the standard basis vectors e1 and e2, just like A.
>
> Since both A and B are diagonal and commute, they share the same set of eigenvectors, which are e1 and e2. However, their eigenvalues are different unless a=1 and d=2, which is not necessarily the case. So, while they share the same eigenvectors, their eigenvalues are distinct unless specified otherwise.
>
> Wait, but the problem just says B commutes with A, not that they have the same eigenvalues. So the conclusion should be that A and B share the same eigenvectors but not necessarily the same eigenvalues. For example, if B is [[3, 0], [0, 4]], it still commutes with A but has different eigenvalues. Therefore, the answer is that they share the same eigenvectors but not necessarily the same eigenvalues.

To determine if matrices A and B share the same eigenvalues and eigenvectors under the condition $AB = BA$:

Eigenvalues:

- Matrix A (diagonal with entries 1 and 2) has eigenvalues 1 and 2.

- Matrix B (also diagonal, of the form \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \)) has eigenvalues **\(a ** and d.
  Unless $a = 1$ and $d = 2$, their eigenvalues are distinct.

Eigenvectors:

- Both A and B are diagonal matrices, so their eigenvectors are the standard basis vectors $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

  These eigenvectors are shared by both matrices.

Conclusion:

- A and B share the same eigenvectors (due to being diagonal and commuting).
- They do not necessarily share the same eigenvalues unless $a = 1$ and $d = 2$.

$$\boxed{\text{They share eigenvectors but not eigenvalues unless } a = 1 \text{ and } d = 2.}$$



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